Online Supplementary Document to "Semiparametric Identification, Estimation and Testing for Endogenous Regressions"

Chaohua Dong*, Jiti Gao†, Oliver Linton⋆ and Bin Peng†

*Zhongnan University of Economics and Law, †Monash University and ⋆University of Cambridge

D Proofs for Appendix A.2

Before proving Theorems A.1–A.4 listed in Appendix A.2, we give one lemma as a preparation.

Lemma D.1. *Under Assumptions A.1 and A.2 and* $\mathbb{E}\left[g^4(\mathbf{x},\boldsymbol{\theta})\right]<\infty$ *for any* $\boldsymbol{\theta}\in\Theta_0$,

$$(i)\ \frac{1}{n}\sum_{i=1}^n \mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)^\top=\mathbb{E}\left[\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)^\top\right]+O_P(n^{-1/2}),$$

$$(ii)\ \frac{1}{n}\sum_{i=1}^n e_i\mathbf{V}_k(\mathbf{x}_i)=O_P(\sqrt{k/n}),$$

$$(iii)\ \frac{1}{n}\mathbf{V}^\top\mathbf{V}=I_k+O_P(kn^{-1/2}),$$

$$(iv)\ \frac{1}{n}\sum_{i=1}^n \mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)\,\mathbf{V}_k(\mathbf{x}_i)^\top=\mathbb{E}\left[\mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)\,\mathbf{V}_k(\mathbf{x})^\top\right]+O_P(\sqrt{k/n}),$$

$$(v)\ \frac{1}{n}\sum_{i=1}^n g(\mathbf{x}_i,\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x}_i)=\mathbb{E}\left[g(\mathbf{x},\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x})\right]+O_P(\sqrt{k/n}),\quad \forall\boldsymbol{\theta}\in\Theta_0;$$

where $\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)=\frac{\partial g(\mathbf{x},\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}$. *When* $g(\mathbf{x},\boldsymbol{\theta})$ *is replaced by* $\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)$ *in (v), the similar assertion also holds.*

Proof of Lemma D.1. (1) Observe that

$$\begin{aligned}&\mathbb{E}\left\|\frac{1}{n}\sum_{i=1}^n \mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)^\top-\mathbb{E}\left[\mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x}_i,\boldsymbol{\theta}_0)^\top\right]\right\|^2\\&=\frac{1}{n^2}\sum_{i=1}^n\mathbb{E}\left\|\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)^\top-\mathbb{E}\left[\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)^\top)\right]\right\|^2\le\frac{1}{n}\mathbb{E}\|\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)\,\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)^\top\|^2\\&=\frac{1}{n}\mathbb{E}\|\mathbf{g}_1(\mathbf{x},\boldsymbol{\theta}_0)\|^4=O(1/n).\end{aligned}$$

(2) Notice that

$$\mathbb{E}\left\|\frac{1}{n}\sum_{i=1}^n e_i\mathbf{V}_k(\mathbf{x}_i)\right\|^2=\frac{\sigma_e^2}{n^2}\sum_{i=1}^n\mathbb{E}\left\|\mathbf{V}_k(\mathbf{x}_i)\right\|^2=\sigma_e^2\,\frac{k}{n}.$$

(3) Also observe that

$$\begin{aligned}\mathbb{E}\left\|\frac{1}{n}\mathbf{V}^\top\mathbf{V}-I_k\right\|^2&=\sum_{j=1}^k\mathbb{E}\left(\frac{1}{n}\sum_{i=1}^n\psi_j(\mathbf{x}_i)^2-1\right)^2+2\sum_{j=2}^k\sum_{\ell=1}^{j-1}\mathbb{E}\left(\frac{1}{n}\sum_{i=1}^n\psi_j(\mathbf{x}_i)\psi_\ell(\mathbf{x}_i)\right)^2\\&=\sum_{j=1}^k\frac{1}{n^2}\sum_{i=1}^n\mathbb{E}\left(\psi_j(\mathbf{x}_i)^2-1\right)^2+2\sum_{j=2}^k\sum_{\ell=1}^{j-1}\frac{1}{n^2}\sum_{i=1}^n\mathbb{E}\left(\psi_j(\mathbf{x}_i)\psi_\ell(\mathbf{x}_i)\right)^2\\&\le\sum_{j=1}^k\frac{1}{n^2}\sum_{i=1}^n\mathbb{E}\psi_j(\mathbf{x}_i)^4+2\sum_{j=2}^k\sum_{\ell=1}^{j-1}\frac{1}{n^2}\sum_{i=1}^n\mathbb{E}\psi_j(\mathbf{x}_i)^2\psi_\ell(\mathbf{x}_i)^2\le C\frac{k^2}{n}.\end{aligned}$$

(4) Note also that

$$\mathbb{E}\left\|\frac{1}{n}\sum_{i=1}^{n}\mathbf{V}_k(\mathbf{x}_i)^\top-\mathbb{E}(\mathbf{x}\mathbf{V}_k(\mathbf{x})^\top)\right\|^2 =\frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\left\|\mathbf{x}_i\mathbf{V}_k(\mathbf{x}_i)^\top-\mathbb{E}(\mathbf{x}\mathbf{V}_k(\mathbf{x})^\top)\right\|^2$$
$$\leq\frac{1}{n}\mathbb{E}\|\mathbf{x}\mathbf{V}_k(\mathbf{x})^\top\|^2\leq C\frac{k}{n}.$$

(5) Finally, we have

$$\mathbb{E}\left\|\frac{1}{n}\sum_{i=1}^{n}g(\mathbf{x}_i,\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x}_i)-\mathbb{E}(g(\mathbf{x},\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x}))\right\|^2 = \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\|g(\mathbf{x}_i,\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x}_i)-\mathbb{E}(g(\mathbf{x},\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x}))\|^2$$
$$\leq\frac{1}{n}\mathbb{E}\|[g(\mathbf{x},\boldsymbol{\theta})\mathbf{V}_k(\mathbf{x})]\|^2\leq C\frac{k}{n}\left(\mathbb{E}[g^4(\mathbf{x},\boldsymbol{\theta})]\right)^{1/2}\sup_j\left(\mathbb{E}[\psi_j^4(\mathbf{x})]\right)^{1/2}=O(k/n).$$

The other proofs follow similarly. □

Proof of Theorem A.1. We shall check a sufficient condition in Park and Phillips (2001, p. 133) to show the consistency. Observe that, as $n\to\infty$, for any $\boldsymbol{\theta}\in\Theta$,

$$\begin{aligned}L_n(\boldsymbol{\theta}) =&\frac{1}{n}\|\mathbf{M}_v(\mathbf{y}-\mathbf{G}(\boldsymbol{\theta}))\|^2=\frac{1}{n}\|\mathbf{M}_v(\mathbf{e}+\boldsymbol{\delta}+\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\|^2\\ =&\frac{1}{n}\|\mathbf{M}_v\mathbf{e}\|^2+\frac{1}{n}\|\mathbf{M}_v\boldsymbol{\delta}\|^2+\frac{1}{n}\|\mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\|^2\\ &+\frac{1}{n}\langle\mathbf{M}_v\mathbf{e},\mathbf{M}_v\boldsymbol{\delta}\rangle+\frac{1}{n}\langle\mathbf{M}_v\mathbf{e},\mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\rangle\\ &+\frac{1}{n}\langle\mathbf{M}_v\boldsymbol{\delta},\mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\rangle.\end{aligned}$$

Observe that

$$\begin{aligned}\frac{1}{n}\|\mathbf{M}_v\mathbf{e}\|^2=&\frac{1}{n}\mathbf{e}^\top\mathbf{e}-\frac{1}{n}\mathbf{e}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{e}\\ =&\frac{1}{n}\sum_{i=1}^{n}e_i^2-\frac{1}{n}\sum_{i=1}^{n}e_i\mathbf{V}_k(\mathbf{x}_i)^\top\left(\frac{1}{n}\mathbf{V}^\top\mathbf{V}\right)^{-1}\frac{1}{n}\sum_{i=1}^{n}e_i\mathbf{V}_k(\mathbf{x}_i),\end{aligned}$$

and by Lemma D.1, we have

$$\frac{1}{n}\sum_{i=1}^{n}e_i\mathbf{V}_k(\mathbf{x}_i)=O_P(\sqrt{k/n})\quad\text{and}\quad\frac{1}{n}\mathbf{V}^\top\mathbf{V}=\mathbf{I}_k+O_P(kn^{-1/2}).$$

This implies $\frac{1}{n}\|\mathbf{M}_v\mathbf{e}\|^2\to_P\sigma_e^2$. Moreover, since $\mathbf{M}_v$ is idempotent,

$$\frac{1}{n}\|\mathbf{M}_v\boldsymbol{\delta}\|^2\leq\frac{1}{n}\|\boldsymbol{\delta}\|^2=\frac{1}{n}\sum_{i=1}^{n}\delta_k^2(\mathbf{x}_i)=O_P(\|\delta_k(\mathbf{x})\|^2)=o_P(1),$$

where $\|\delta_k(\mathbf{x})\|^2=\sum_{j=k+1}^{\infty}\gamma_j^2\to 0$ as $k\to\infty$. And, we have

$$\begin{aligned}&\frac{1}{n}\langle\mathbf{M}_v\mathbf{e},\mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\rangle=\frac{1}{n}\mathbf{e}^\top\mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\\ =&\frac{1}{n}\mathbf{e}^\top(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))-\frac{1}{n}\mathbf{e}^\top\mathbf{V}\left(\mathbf{V}^\top\mathbf{V}\right)^{-1}\mathbf{V}^\top(\mathbf{G}(\boldsymbol{\theta}_0)-\mathbf{G}(\boldsymbol{\theta}))\\ =&\frac{1}{n}\sum_{i=1}^{n}(g(\mathbf{x}_i,\boldsymbol{\theta})-g(\mathbf{x}_i,\boldsymbol{\theta}_0))e_i\\ &-\frac{1}{n}\sum_{i=1}^{n}e_i\mathbf{V}_k(\mathbf{x}_i)^\top\left(\frac{1}{n}\mathbf{V}^\top\mathbf{V}\right)^{-1}\frac{1}{n}\sum_{i=1}^{n}(g(\mathbf{x}_i,\boldsymbol{\theta})-g(\mathbf{x}_i,\boldsymbol{\theta}_0))\mathbf{V}_k(\mathbf{x}_i)\end{aligned}$$

$$=O_P(n^{-1/2}) + O_P(k/n) = o_P(1).$$

Furthermore, we obtain

$$\begin{aligned}
&\frac{1}{n}\|\mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0) - \mathbf{G}(\boldsymbol{\theta}))\|^2 = \frac{1}{n}(\mathbf{G}(\boldsymbol{\theta}_0) - \mathbf{G}(\boldsymbol{\theta}))^\top \mathbf{M}_v(\mathbf{G}(\boldsymbol{\theta}_0) - \mathbf{G}(\boldsymbol{\theta})) \\
= \;& \frac{1}{n}(\mathbf{G}(\boldsymbol{\theta}_0) - \mathbf{G}(\boldsymbol{\theta}))^\top (G(\boldsymbol{\theta}_0) - G(\boldsymbol{\theta})) - \frac{1}{n}(\mathbf{G}(\boldsymbol{\theta}_0) - \mathbf{G}(\boldsymbol{\theta}))^\top \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top(\mathbf{G}(\boldsymbol{\theta}_0) - \mathbf{G}(\boldsymbol{\theta})) \\
= \;& \frac{1}{n}\sum_{i=1}^n (g(\mathbf{x}_i, \boldsymbol{\theta}) - g(\mathbf{x}_i, \boldsymbol{\theta}_0))^2 \\
& - \frac{1}{n}\sum_{i=1}^n (g(\mathbf{x}_i, \boldsymbol{\theta}) - g(\mathbf{x}_i, \boldsymbol{\theta}_0))\mathbf{V}_k(\mathbf{x}_i)^\top \left(\frac{1}{n}\mathbf{V}^\top\mathbf{V}\right)^{-1} \frac{1}{n}\sum_{i=1}^n (g(\mathbf{x}_i, \boldsymbol{\theta}) - g(\mathbf{x}_i, \boldsymbol{\theta}_0))\mathbf{V}_k(\mathbf{x}_i) \\
= \;& \mathbb{E}[(g(\mathbf{x}, \boldsymbol{\theta}) - g(\mathbf{x}, \boldsymbol{\theta}_0))^2] + \mathbb{E}[(g(\mathbf{x}, \boldsymbol{\theta}) - g(\mathbf{x}, \boldsymbol{\theta}_0))\mathbf{V}_k(\mathbf{x})^\top]\mathbb{E}[(g(\mathbf{x}, \boldsymbol{\theta}) - g(\mathbf{x}, \boldsymbol{\theta}_0))\mathbf{V}_k(\mathbf{x})] + o_P(1) \\
\to_P \;& \mathbb{E}[(g(\mathbf{x}, \boldsymbol{\theta}) - g(\mathbf{x}, \boldsymbol{\theta}_0))^2] - \sum_{j=1}^\infty \{\mathbb{E}[(g(\mathbf{x}, \boldsymbol{\theta}) - g(\mathbf{x}, \boldsymbol{\theta}_0))\psi_j(\mathbf{x})]\}^2 \equiv: M(\boldsymbol{\theta}, \boldsymbol{\theta}_0).
\end{aligned}$$

Finally, we have

$$L_n(\boldsymbol{\theta}) \to_P \sigma_e^2 + M(\boldsymbol{\theta}, \boldsymbol{\theta}_0) \equiv: L(\boldsymbol{\theta}, \boldsymbol{\theta}_0).$$

It is clear from Assumption 3.1(iii) that when $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, $L(\boldsymbol{\theta}, \boldsymbol{\theta}_0)$ achieves its minimum, σ_e^2.

Conversely, when $L(\boldsymbol{\theta}, \boldsymbol{\theta}_0)$ achieves its minimum we have $M\boldsymbol{\theta}, \boldsymbol{\theta}_0) = 0$, which implies $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ by the conditions listed in the theorem. The asymptotic consistency holds. □

Proof of Theorem A.2. Note that

$$\begin{aligned}
\sqrt{n}\mathbf{S}_n(\boldsymbol{\theta}_0) &= \frac{1}{\sqrt{n}}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v(\mathbf{y} - \mathbf{G}(\boldsymbol{\theta}_0)) = \frac{1}{\sqrt{n}}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v(\mathbf{e} + \boldsymbol{\delta}) \\
&= \frac{1}{\sqrt{n}}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v\mathbf{e} + \frac{1}{\sqrt{n}}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v\boldsymbol{\delta}. \qquad \text{(D.1)}
\end{aligned}$$

We start to deal with the last term of equation (D.1) as follows:

$$\frac{1}{\sqrt{n}}\left\|\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}^\top \mathbf{M}_v\boldsymbol{\delta}\right\| \le \frac{1}{\sqrt{n}}\left\|\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}\right\| \|\boldsymbol{\delta}\| = \sqrt{\frac{1}{n}\sum_{i=1}^n \|\mathbf{g}_1(\mathbf{x}_i, \boldsymbol{\theta}_0)\|^2}\sqrt{\sum_{i=1}^n |\delta_k(\mathbf{x}_i)|^2}, \qquad \text{(D.2)}$$

and $\mathbb{E}[\delta_k^2(\mathbf{x}_i)] = \int_{\mathcal{X}} \delta_k^2(\mathbf{x}) f(\mathbf{x}) d\mathbf{x} = o(k^{-2s/d})$ by Lemma A.5 in Dong et al. (2021) (see also Theorem C.4.13 of Dong and Gao (2025)). This implies $\frac{1}{\sqrt{n}}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}^\top \mathbf{M}_v\boldsymbol{\delta} = O_P(\sqrt{n}\, k^{-2s/d}) = o_P(1)$ by Assumption 3.1.

It is clear that the first term has expectation zero, and

$$\begin{aligned}
\frac{1}{\sqrt{n}}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v\mathbf{e} =& \frac{1}{\sqrt{n}}\left[\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top - \frac{1}{n}\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{V}\left(\frac{1}{n}\mathbf{V}^\top\mathbf{V}\right)^{-1}\mathbf{V}^\top\right]\mathbf{e} \\
=& \frac{1}{\sqrt{n}}\left[\frac{\partial}{\partial\boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top - \frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x}_i)^\top\left(\frac{1}{n}\mathbf{V}^\top\mathbf{V}\right)^{-1}\mathbf{V}^\top\right]\mathbf{e} \\
=& \frac{1}{\sqrt{n}}\left[\frac{\partial}{\partial\boldsymbol{\theta}}G(\boldsymbol{\theta}_0)^\top - \left\{\mathbb{E}\left(\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\right\}\mathbf{V}^\top\right]\mathbf{e}\,(1 + o_P(1)) \\
=& \frac{1}{\sqrt{n}}\sum_{i=1}^n\left[\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i, \boldsymbol{\theta}_0) - \left\{\mathbb{E}\left(\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\right\}\mathbf{V}_k(\mathbf{x}_i)\right] e_i \;(1 + o_P(1)).
\end{aligned}$$

Since the data is of i.i.d. sequence and the dimension is fixed, to show the limit distribution of the score function, we only need to calculate the limit of the covariance matrix. In fact,

$$\frac{1}{n}\mathbb{E}\left(\sum_{i=1}^n\left[\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i, \boldsymbol{\theta}_0) - \left\{\mathbb{E}\left(\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\right\}\mathbf{V}_k(\mathbf{x}_i)\right] e_i\right.$$

$$
\begin{aligned}
&\times \sum_{i=1}^{n} \left[\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) - \left\{\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\right\} \mathbf{V}_k(\mathbf{x}_i)\right]^\top e_i\Bigg) \\
=&\frac{\sigma_e^2}{n} \sum_{i=1}^{n} \mathbb{E}\Bigg(\left[\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) - \left\{\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\right\} \mathbf{V}_k(\mathbf{x}_i)\right] \\
&\times \left[\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) - \left\{\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\right\} \mathbf{V}_k(\mathbf{x}_i)\right]^\top\Bigg) \\
=&\sigma_e^2 \left[\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)\right) - \mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x})^\top\right)\mathbb{E}\left(\mathbf{V}_k(\mathbf{x})\frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)\right)\right] \\
=&\sigma_e^2 \left[\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)\right) - \sum_{j=1}^{k}\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\psi_j(\mathbf{x})\right)\mathbb{E}\left(\psi_j(\mathbf{x})\frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)\right)\right] \\
\to&\sigma_e^2 \left[\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)\right) - \sum_{j=1}^{\infty}\mathbb{E}\left(\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\psi_j(\mathbf{x})\right)\mathbb{E}\left(\psi_j(\mathbf{x})\frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)\right)\right] \\
\equiv:&\sigma_e^2 \boldsymbol{\Sigma}_g.
\end{aligned}
$$

Hence, $\sqrt{n}\,\mathbf{S}_n(\boldsymbol{\theta}_0) \to_{\mathcal{D}} N(0, \sigma_e^2 \boldsymbol{\Sigma}_g)$ as $n \to \infty$. Next, consider the Hessian matrix:

$$
\begin{aligned}
\mathbf{H}_n(\boldsymbol{\theta}_0) =&\frac{1}{n}\frac{\partial}{\partial \boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v \frac{\partial}{\partial \boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0) - \frac{1}{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\mathbf{G}(\boldsymbol{\theta}_0)\mathbf{M}_v(\mathbf{y} - \mathbf{G}(\boldsymbol{\theta}_0)) \\
=&\frac{1}{n}\frac{\partial}{\partial \boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v \frac{\partial}{\partial \boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0) - \frac{1}{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\mathbf{G}(\boldsymbol{\theta}_0)\mathbf{M}_v(\mathbf{e} + \boldsymbol{\delta}).
\end{aligned}
$$

As shown in the convergence of $\mathbf{S}_n(\boldsymbol{\theta}_0)$,

$$
\frac{1}{n}\frac{\partial}{\partial \boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0)^\top \mathbf{M}_v \frac{\partial}{\partial \boldsymbol{\theta}}\mathbf{G}(\boldsymbol{\theta}_0) \to_P \boldsymbol{\Sigma}_g \quad \text{and} \quad \frac{1}{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\mathbf{G}(\boldsymbol{\theta}_0)\mathbf{M}_v\mathbf{e} = O_P(n^{-1/2}).
$$

To fulfil the convergence of $\mathbf{H}_n(\boldsymbol{\theta}_0)$, we need only to prove the negligibility of the last term. Indeed,

$$
\begin{aligned}
&\frac{1}{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\mathbf{G}(\boldsymbol{\theta}_0)\mathbf{M}_v\boldsymbol{\delta} = \frac{1}{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\mathbf{G}(\boldsymbol{\theta}_0)\boldsymbol{\delta} - \frac{1}{n^2}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\mathbf{G}(\boldsymbol{\theta}_0)\mathbf{V}\mathbf{V}^\top\boldsymbol{\delta} \\
=& \frac{1}{n}\sum_{i=1}^{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}_i, \boldsymbol{\theta}_0)\delta_k(\mathbf{x}_i) - \frac{1}{n}\sum_{i=1}^{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}_i, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x}_i)^\top\frac{1}{n}\sum_{i=1}^{n}\mathbf{V}_k(\mathbf{x}_i)\delta_k(\mathbf{x}_i),
\end{aligned} \tag{D.3}
$$

where

$$
\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left\|\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}_i, \boldsymbol{\theta}_0)\delta_k(\mathbf{x}_i)\right\| \le \left(\mathbb{E}\left\|\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}, \boldsymbol{\theta}_0)\right\|^2\mathbb{E}\left[\delta_k^2(\mathbf{x})\right]\right)^{1/2} = O(\|\delta_k(\mathbf{x})\|) = o(1), \tag{D.4}
$$

$$
\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left\|\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}_i, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x}_i)^\top\right\| \le \left(\mathbb{E}\left\|\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}, \boldsymbol{\theta}_0)\right\|^2\mathbb{E}[\|\mathbf{V}_k(\mathbf{x})\|^2]\right)^{1/2} = O(\sqrt{k}), \tag{D.5}
$$

$$
\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\|\mathbf{V}_k(\mathbf{x}_i)\delta_k(\mathbf{x}_i)\| \le \left(\mathbb{E}\|\mathbf{V}_k(\mathbf{x})\|^2\mathbb{E}[\delta_k^2(\mathbf{x})]\right)^{1/2} = O(\sqrt{k}\|\delta_k(\mathbf{x})\|). \tag{D.6}
$$

Therefore, in the same way as in the derivation of (D.2), equations (D.5) and (D.6) imply that the last term of (D.3) becomes

$$
\left\|\frac{1}{n}\sum_{i=1}^{n}\frac{\partial^2}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}g(\mathbf{x}_i, \boldsymbol{\theta}_0)\mathbf{V}_k(\mathbf{x}_i)^\top\frac{1}{n}\sum_{i=1}^{n}\mathbf{V}_k(\mathbf{x}_i)\delta_k(\mathbf{x}_i)\right\| = O_P\left(k\,\|\delta_k(\mathbf{x})\|\right) = o_P\left(k^{\frac{d-s}{d}}\right) = o_P(1), \tag{D.7}
$$

using Assumption 3.1 for the case of $s \ge d$, and accordingly, $H_n(\boldsymbol{\theta}_0) \to_P \boldsymbol{\Sigma}_g$ as $n \to \infty$.

By the first order condition of (A.14), $S_n(\widehat{\boldsymbol{\theta}}) = 0$. Due to the consistency,

$$
\begin{aligned}
0 = \mathbf{S}_n(\widehat{\boldsymbol{\theta}}) =& \mathbf{S}_n(\boldsymbol{\theta}_0) + \mathbf{H}_n(\tilde{\boldsymbol{\theta}})(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \\
=& \mathbf{S}_n(\boldsymbol{\theta}_0) + \mathbf{H}_n(\boldsymbol{\theta}_0)(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + [\mathbf{H}_n(\tilde{\boldsymbol{\theta}}) - \mathbf{H}_n(\boldsymbol{\theta}_0)](\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0),
\end{aligned}
$$

where $\tilde{\boldsymbol{\theta}}$ is on the line jointing $\widehat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$, implying $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| \leq \|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|$.

By Assumption 3.1, we may write

$$
\begin{aligned}
0 &= \sqrt{n}\mathbf{S}_n(\boldsymbol{\theta}_0) + \mathbf{H}_n(\boldsymbol{\theta}_0)\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + [\mathbf{H}_n(\tilde{\boldsymbol{\theta}}) - \mathbf{H}_n(\boldsymbol{\theta}_0)]\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \\
&= \sqrt{n}\,\mathbf{S}_n(\boldsymbol{\theta}_0) + \mathbf{H}_n(\boldsymbol{\theta}_0)\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)(1 + o_P(1)).
\end{aligned}
$$

Hence, we have as $n \to \infty$, $\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = [\mathbf{H}_n(\boldsymbol{\theta}_0)]^{-1}\sqrt{n}\,\mathbf{S}_n(\boldsymbol{\theta}_0)(1 + o_P(1)) \to_{\mathcal{D}} N(0, \sigma_e^2\,\boldsymbol{\Sigma}_g^{-1})$ using the conventional continuous mapping theorem and Slutsky theorem. □

Proof of Theorem A.3. Notice that, for $i \neq j$, by the mutual independence of $\mathbf{x}_i$ and $\mathbf{x}_j$ and the orthogonality of $\{\psi_\ell(x)\}$ we have

$$
\begin{aligned}
&\mathbb{E}\left[\left(\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\right)^2\right] \\
=& \sum_{k=k_{\min}}^{k_{\max}} \mathbb{E}[\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)]^2 + \sum_{k=k_{\min}}^{k_{\max}} \sum_{p=k_{\min},\neq k}^{k_{\max}} \mathbb{E}[\mathbf{V}_p^\top(\mathbf{x}_i)\mathbf{V}_p(\mathbf{x}_j)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)] \\
=& \sum_{k=k_{\min}}^{k_{\max}} \mathbb{E}\mathrm{tr}[\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)]^2 + \sum_{k=k_{\min}}^{k_{\max}} \sum_{p=k_{\min},\neq k}^{k_{\max}} \mathbb{E}\mathrm{tr}[\mathbf{V}_p^\top(\mathbf{x}_i)\mathbf{V}_p(\mathbf{x}_j)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)] \\
=& \sum_{k=k_{\min}}^{k_{\max}} \mathrm{tr}\{[\mathbb{E}\mathbf{V}_k(\mathbf{x}_i)\mathbf{V}_k^\top(\mathbf{x}_i)][\mathbb{E}\mathbf{V}_k(\mathbf{x}_j)\mathbf{V}_k^\top(\mathbf{x}_j)]\} + \sum_{k=k_{\min}}^{k_{\max}} \sum_{p=k_{\min},\neq k}^{k_{\max}} \mathrm{tr}\{[\mathbb{E}\mathbf{V}_p(\mathbf{x}_j)\mathbf{V}_k^\top(\mathbf{x}_j)][\mathbb{E}\mathbf{V}_k(\mathbf{x}_i)\mathbf{V}_p^\top(\mathbf{x}_i)]\} \\
=& \sum_{k=k_{\min}}^{k_{\max}} k + \sum_{k=k_{\min}}^{k_{\max}} \sum_{p=k_{\min},\neq k}^{k_{\max}} \min(k,p) = \frac{1}{2}(k_{\max} + k_{\min})(k_{\max} - k_{\min} + 1) + 2\sum_{k=k_{\min}+1}^{k_{\max}} \sum_{p=k_{\min}}^{k-1} p \\
=& \frac{1}{2}(k_{\max} + k_{\min})(k_{\max} - k_{\min} + 1) + \sum_{k=k_{\min}+1}^{k_{\max}} (k + k_{\min} - 1)(k - k_{\min}) \\
=& \frac{1}{2}(k_{\max} + k_{\min})(k_{\max} - k_{\min} + 1) + \sum_{k=k_{\min}+1}^{k_{\max}} k^2 - k_{\min}^2(k_{\max} - k_{\min} - 1) - \sum_{k=k_{\min}+1}^{k_{\max}} (k - k_{\min}) \\
=& \frac{k_{\max}(k_{\max}+1)(2k_{\max}+1) - (k_{\min}+1)(k_{\min}+2)(2k_{\min}+3)}{6} \\
&- k_{\min}^2(k_{\max} - k_{\min} - 1) + k_{\min}(k_{\max} - k_{\min}) \\
=& \frac{1}{3}k_{\max}^3(1 + o(1)).
\end{aligned}
$$

We then consider the convergence of $\widehat{\sigma}_e^2 \to_P \sigma_\varepsilon^2 = \mathbb{E}[\varepsilon^2]$ as $n \to \infty$. Under H_0, $m_n(\mathbf{x}) = 0$ almost surely, and the model $y = g(\mathbf{x}, \boldsymbol{\theta}_0) + \varepsilon$ satisfies $\mathbb{E}[\varepsilon|\mathbf{x}] = 0$. Moreover,

$$
\widetilde{e}_i = y_i - g(\mathbf{x}, \widehat{\boldsymbol{\theta}}) = g(\mathbf{x}, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) + \varepsilon_i, \quad i = 1, \cdots, n.
$$

It follows that

$$
\begin{aligned}
\widehat{\sigma}_e^2 =& \frac{1}{n}\sum_{i=1}^{n} \widetilde{e}_i^2 = \frac{1}{n}\sum_{i=1}^{n}[g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) + \varepsilon_i]^2 \\
=& \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i^2 + \frac{1}{n}\sum_{i=1}^{n}[g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})]^2 + \frac{2}{n}\sum_{i=1}^{n}[g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})]\varepsilon_i
\end{aligned}
$$

$$
\begin{aligned}
=&\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2+(\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)^\top\left[\frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\right](\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)\\
&-(\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)^\top\frac{2}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\varepsilon_i,
\end{aligned}
$$

where we use the first order Taylor approximation for $g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})$ due to the consistency of $\widehat{\boldsymbol{\theta}}$.

It is clear that due to the weak LLN,

$$
\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2\to_P\sigma_\varepsilon^2,
$$

as $n\to\infty$; and the second and third terms are $o_P(1)$ due to the i.i.d. data and the consistency of $\widehat{\boldsymbol{\theta}}$.

This gives $\widehat{\sigma}_e^2\to_P\sigma_\varepsilon^2$ as $n\to\infty$. Hence, this allows us to replace $\widehat{\sigma}_e^4$ by σ_ε^4 due to the continuous mapping theorem when we prove the normality of T_n/σ_n. We now conclude $\sigma_n=O_P(nk_{\max}^{3/2})$ that is used in the sequel.

Notice that

$$
\begin{aligned}
T_n=&\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\widetilde{e}_i\,\widetilde{e}_j\\
=&\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})+\varepsilon_j][g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})+\varepsilon_i]\\
=&\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\varepsilon_j\varepsilon_i\\
&+\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})][g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})]\\
&+2\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})]\varepsilon_i\equiv:T_{1n}+T_{2n}+T_{3n}.
\end{aligned}
$$

Then, we shall show, as $n\to\infty$,

$$
\frac{1}{\sigma_n}T_{1n}=\frac{2}{\sigma_n}\sum_{i=2}^{n}\left[\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\varepsilon_j\right]\varepsilon_i\to_{\mathcal{D}}N(0,1),
$$

and $\frac{1}{\sigma_n}T_{in}=o_P(1),i=2,3$.

Because $\mathbb{E}[\varepsilon_i|\mathbf{x}_i]=0$, and $\{\mathbf{x}_i,\varepsilon_i\}$ are mutually independent over i, one may construct a filtration $\mathcal{F}_i=\sigma(\mathbf{x}_s,s\le i;\varepsilon_t,t<i)$ such that $(\varepsilon_i,\mathcal{F}_i)$ is a martingale difference sequence. Thus, to show $T_{1n}/\sigma_n\to_{\mathcal{D}}N(0,1)$, it suffices to check whether it satisfies the convergence of the conditional variance process and the Linderberg condition in Corollary 3.1 of Hall and Heyde (1980).

Towards this, denote $a_{ij}\equiv\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)$ for convenience. The conditional variance process of T_{1n}/σ_n is

$$
\begin{aligned}
\frac{4\sigma_\varepsilon^2}{\sigma_n^2}\sum_{i=2}^{n}\left[\sum_{j=1}^{i-1}a_{ij}\varepsilon_j\right]^2=&\frac{4\sigma_\varepsilon^2}{\sigma_n^2}\sum_{i=2}^{n}\sum_{j=1}^{i-1}a_{ij}^2\varepsilon_j^2+\frac{8\sigma_\varepsilon^2}{\sigma_n^2}\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}a_{ij_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{ij_2}\varepsilon_{j_2}\\
=&\frac{4\sigma_\varepsilon^4}{\sigma_n^2}\sum_{i=2}^{n}\sum_{j=1}^{i-1}a_{ij}^2+\frac{4\sigma_\varepsilon^2}{\sigma_n^2}\sum_{i=2}^{n}\sum_{j=1}^{i-1}a_{ij}^2(\varepsilon_j^2-\sigma_\varepsilon^2)\\
&+\frac{8\sigma_\varepsilon^2}{\sigma_n^2}\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}a_{ij_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{ij_2}\varepsilon_{j_2}=I_{1n}+I_{2n}+I_{3n},\quad\text{say.}
\end{aligned}
$$

We shall show $I_{1n} \to_P 1$, $I_{2n} = o_P(1)$ and $I_{3n} = o_P(1)$. To begin, by the definition of σ_n^2,

$$\begin{aligned}
\mathbb{E}(I_{1n}-1)^2 =& \frac{16}{n^4 k_{\max}^6}\mathbb{E}\left[\sum_{i=2}^{n}\sum_{j=1}^{i-1}(a_{ij}^2 - \mathbb{E}a_{ij}^2)\right]^2 = \frac{16}{n^4 k_{\max}^6}\sum_{i=2}^{n}\mathbb{E}\left[\sum_{j=1}^{i-1}(a_{ij}^2 - \mathbb{E}a_{ij}^2)\right]^2 \\
&+ \frac{32}{n^4 k_{\max}^6}\mathbb{E}\sum_{i_1=3}^{n}\sum_{i_2=2}^{i_1-1}\sum_{j=1}^{i_1-1}(a_{i_1 j}^2 - \mathbb{E}a_{i_1 j}^2)\sum_{j=1}^{i_2-1}(a_{i_2 j}^2 - \mathbb{E}a_{i_2 j}^2) \\
=& \frac{16}{n^4 k_{\max}^6}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\mathbb{E}\left[(a_{ij}^2 - \mathbb{E}a_{ij}^2)\right]^2 \\
&+ \frac{32}{n^4 k_{\max}^6}\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}\sum_{j_2=1}^{j_1-1}\mathbb{E}\left[(a_{ij_1}^2 - \mathbb{E}a_{ij_1}^2)(a_{ij_2}^2 - \mathbb{E}a_{ij_2}^2)\right] \\
&+ \frac{32}{n^4 k_{\max}^6}\sum_{i_1=3}^{n}\sum_{i_2=2}^{i_1-1}\sum_{j_1=1}^{i_1-1}\sum_{j_2=1}^{i_2-1}\mathbb{E}[(a_{i_1 j_1}^2 - \mathbb{E}a_{i_1 j_1}^2)(a_{i_2 j_2}^2 - \mathbb{E}a_{i_2 j_2}^2)] \\
\le& \frac{C}{n^2}k_{\max}^2 + \frac{C}{n}k_{\max}^2 = o(1),
\end{aligned}$$

in view of $\frac{k_{\max}^2}{n} \to 0$, because straightforward algebra yields $\mathbb{E}[a_{ij}^4] \le Ck_{\max}^8$ by the definition of $\mathbf{V}_k(\cdot)$; similarly, $|\mathbb{E}(a_{ij_1}^2 - \mathbb{E}[a_{ij_1}^2])(a_{ij_2}^2 - \mathbb{E}[a_{ij_2}^2])| \le \mathbb{E}|a_{ij_1}^2 a_{ij_2}^2| \le Ck_{\max}^8$; and

$$\mathbb{E}[(a_{i_1 j_1}^2 - \mathbb{E}[a_{i_1 j_1}^2])(a_{i_2 j_2}^2 - \mathbb{E}[a_{i_2 j_2}^2])] = \mathbb{E}[(a_{i_1 j_1}^2 - \mathbb{E}a_{i_1 j_1}^2)]\mathbb{E}[(a_{i_2 j_2}^2 - \mathbb{E}[a_{i_2 j_2}^2])] = 0,$$

due to independence of the data. This implies that $I_{1n} \to_P 1$ as $n \to \infty$.

We then consider I_{2n}. Observe that

$$\begin{aligned}
\mathbb{E}(I_{2n}^2) =& \frac{16}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\mathbb{E}\left[\sum_{i=2}^{n}\sum_{j=1}^{i-1}a_{ij}^2(\varepsilon_j^2 - \sigma_\varepsilon^2)\right]^2 = \frac{16}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i=2}^{n}\mathbb{E}\left[\sum_{j=1}^{i-1}a_{ij}^2(\varepsilon_j^2 - \sigma_\varepsilon^2)\right]^2 \\
&+ \frac{32}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i_1=3}^{n}\sum_{i_2=2}^{i_1-1}\sum_{j_1=1}^{i_1-1}\sum_{j_2=1}^{i_2-1}\mathbb{E}[a_{i_1 j_1}^2(\varepsilon_{j_1}^2 - \sigma_\varepsilon^2)a_{i_2 j_2}^2(\varepsilon_{j_2}^2 - \sigma_\varepsilon^2)] \\
=& \frac{16}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\mathbb{E}\left[a_{ij}^4(\varepsilon_j^2 - \sigma_\varepsilon^2)^2\right] \\
&+ \frac{32}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i=2}^{n}\sum_{j_1=2}^{i-1}\sum_{j_2=1}^{j_1-1}\mathbb{E}\left[a_{ij_1}^2 a_{ij_2}^2(\varepsilon_{j_1}^2 - \sigma_\varepsilon^2)(\varepsilon_{j_2}^2 - \sigma_\varepsilon^2)\right] \\
&+ \frac{32}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i_1=3}^{n}\sum_{i_2=2}^{i_1-1}\sum_{j_1=1}^{i_1-1}\sum_{j_2=1}^{i_2-1}\mathbb{E}[a_{i_1 j_1}^2(\varepsilon_{j_1}^2 - \sigma_\varepsilon^2)a_{i_2 j_2}^2(\varepsilon_{j_2}^2 - \sigma_\varepsilon^2)] \\
\le& \frac{C}{n^2}k_{\max}^2 + \frac{32}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i_1=3}^{n}\sum_{i_2=2}^{i_1-1}\sum_{j_2=1}^{i_2-1}\mathbb{E}[a_{i_1 j_2}^2 a_{i_2 j_2}^2(\varepsilon_{j_2}^2 - \sigma_\varepsilon^2)^2] \\
\le& \frac{C}{n^2}k_{\max}^2 + \frac{C}{n}k_{\max}^2 = o(1),
\end{aligned}$$

by $\frac{k_{\max}^2}{n} = o(1)$. For I_{3n}, note that

$$\begin{aligned}
\mathbb{E}(I_{3n}^2) =& \frac{64}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\mathbb{E}\left[\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}a_{ij_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{ij_2}\varepsilon_{j_2}\right]^2 = \frac{64}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i=3}^{n}\mathbb{E}\left[\sum_{j_1=2}^{i-1}a_{ij_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{ij_2}\varepsilon_{j_2}\right]^2 \\
&+ \frac{128}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i_1=4}^{n}\sum_{i_2=3}^{i_1-1}\mathbb{E}\left[\sum_{j_1=2}^{i_1-1}a_{i_1 j_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{i_1 j_2}\varepsilon_{j_2}\right]\left[\sum_{j_1=2}^{i_2-1}a_{i_2 j_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{i_2 j_2}\varepsilon_{j_2}\right]
\end{aligned}$$

$$
\begin{aligned}
=&\frac{64}{\sigma_\varepsilon^4 n^4 k_{\max}^6}\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}\mathbb{E}\left[a_{ij_1}\varepsilon_{j_1}\sum_{j_2=1}^{j_1-1}a_{ij_2}\varepsilon_{j_2}\right]^2+\frac{128}{n^4k_{\max}^6}\sum_{i_1=4}^{n}\sum_{i_2=3}^{i_1-1}\sum_{j_1=2}^{i_2-1}\sum_{j_2=1}^{j_1-1}\mathbb{E}\left[a_{i_1j_1}a_{i_2j_1}a_{i_1j_2}a_{i_2j_2}\right]\\
=&\frac{64}{n^4k_{\max}^6}\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}\sum_{j_2=1}^{j_1-1}\mathbb{E}[a_{ij_1}^2a_{ij_2}^2]+\frac{128}{n^4k_{\max}^6}\sum_{i_1=4}^{n}\sum_{i_2=3}^{i_1-1}\sum_{j_1=2}^{i_2-1}\sum_{j_2=1}^{j_1-1}\sum_{k=k_{\min}}^{k_{\max}}\\
&\quad\mathbb{E}[\mathbf{V}_k^\top(\mathbf{x}_{i_1})\mathbf{V}_k(\mathbf{x}_{j_1})\mathbf{V}_k^\top(\mathbf{x}_{i_2})\mathbf{V}_k(\mathbf{x}_{j_1})\mathbf{V}_k^\top(\mathbf{x}_{i_1})\mathbf{V}_k(\mathbf{x}_{j_2})\mathbf{V}_k^\top(\mathbf{x}_{i_2})\mathbf{V}_k(\mathbf{x}_{j_2})]\\
&+\frac{128}{n^4k_{\max}^6}\sum_{i_1=4}^{n}\sum_{i_2=3}^{i_1-1}\sum_{j_1=2}^{i_2-1}\sum_{j_2=1}^{j_1-1}\mathbb{E}\left[\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_{i_1})\mathbf{V}_k(\mathbf{x}_{j_1})\right]^2\mathbb{E}\left[\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_{i_2})\mathbf{V}_k(\mathbf{x}_{j_2})\right]^2\\
\le&\frac{C}{n}k_{\max}^2+\frac{C}{k_{\max}}+\frac{128}{n^4k_{\max}^6}\sum_{i_1=4}^{n}\sum_{i_2=3}^{i_1-1}\sum_{j_1=2}^{i_2-1}\sum_{j_2=1}^{j_1-1}\\
&\quad\sum_{k=k_{\min}}^{k_{\max}}\mathbb{E}\left[\mathbf{V}_k^\top(\mathbf{x}_{i_1})\mathbf{V}_k(\mathbf{x}_{j_1})\right]^2\sum_{k=k_{\min}}^{k_{\max}}\mathbb{E}\left[\mathbf{V}_k^\top(\mathbf{x}_{i_2})\mathbf{V}_k(\mathbf{x}_{j_2})\right]^2\\
=&\frac{C}{n}k_{\max}^2+\frac{C}{k_{\max}}+\frac{C}{k_{\max}^2}=o(1),
\end{aligned}
$$

where we use the i.i.d. property of the data, and $\mathbb{E}[\psi_j(\mathbf{x})]=0$.

We finish the proof of conditional covariance process of T_{1n}/σ_n converging to 1. We are then about to show that the Lindeberg condition is fulfilled. In doing so, it suffices to prove, denoting $\mu_4=\mathbb{E}[\varepsilon^4]$,

$$
\begin{aligned}
&16\mu_4\frac{1}{\sigma_n^4}\sum_{i=2}^{n}\mathbb{E}\left[\sum_{j=1}^{i-1}\left(\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_j)\mathbf{V}_k(\mathbf{x}_i)\right)\varepsilon_j\right]^4\\
=&16\mu_4^2\frac{1}{n^4k_{\max}^6}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\mathbb{E}\left[\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_j)\mathbf{V}_k(\mathbf{x}_i)\right]^4\\
&+16\mu_4\frac{6\sigma_\varepsilon^4}{n^4k_{\max}^6}\sum_{i=3}^{n}\sum_{j_1=2}^{i-1}\sum_{j_2=1}^{j_1-1}\mathbb{E}\left[\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_{j_1})\mathbf{V}_k(\mathbf{x}_i)\right]^2\left[\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_{j_2})\mathbf{V}_k(\mathbf{x}_i)\right]^2\\
\le&\frac{C}{n^2}k_{\max}^2+\frac{C}{n}k_{\max}^2=o(1).
\end{aligned}
$$

Therefore, invoking Corollary 3.1 of Hall and Heyde (1980), $T_{1n}/\sigma_n\to_{\mathcal{D}}N(0,1)$ as $n\to\infty$.

In what follows, we show $T_{2n}/\sigma_n=o_P(1)$ and $T_{3n}/\sigma_n=o_P(1)$. Notice that $\sigma_n^2=O_P(n^2k_{\max}^3)$, and then

$$
\begin{aligned}
\frac{1}{\sigma_n}T_{2n}=&\frac{1}{\sigma_n}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})][g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})]\\
=&(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})^\top\left[\frac{1}{\sigma_n}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right](\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})\\
=&\sigma_e^4\sqrt{n}(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})^\top\left[\frac{1}{n^2k_{\max}^{3/2}}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right]\sqrt{n}(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}}).
\end{aligned}
$$

Once we show the term in the square bracket is $o_P(1)$, the assertion shall be fulfilled. Observe that

$$
\begin{aligned}
&\frac{1}{n^2k_{\max}^{3/2}}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\\
=&\frac{1}{n^2k_{\max}^{3/2}}\left[\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=k_{\min}}^{k_{\max}}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right.
\end{aligned}
$$

$$
- \sum_{i=1}^{n} \sum_{k=k_{\min}}^{k_{\max}} \|\mathbf{V}_k(\mathbf{x}_i)\|^2 \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \Bigg] \equiv I_1 - I_2, \quad \text{say}.
$$

For I_1, notice that

$$
\begin{aligned}
I_1 =& \frac{1}{n^2 k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \sum_{i=1}^{n} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x}_i) \sum_{j=1}^{n} \mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0) \\
=& \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right] \left[\mathbb{E} \mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0)\right] \\
&+ \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right] \left[\mathbb{E} \mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0)\right] \\
&+ \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right] \left[\frac{1}{n} \sum_{j=1}^{n} \mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0) - \mathbb{E} \mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0)\right] \\
&+ \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right] \\
&\quad \times \left[\frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right] \\
\equiv& I_{11} + I_{12} + I_{13} + I_{14}, \quad \text{say}.
\end{aligned}
$$

It suffices to show $I_{11} = o(1)$, $I_{12} = o_P(1)$, since I_{13} has the same order as I_{12} while I_{14} is of smaller order. In fact,

$$
\|I_{11}\| \le \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left\|\mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right\|^2 \le \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left\|\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)\right\|^2 = O(k_{\max}^{-1/2}) = o(1),
$$

where the second inequality is because $\mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})$ is the coefficients of $\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0)$ projecting on the space generated by $\mathbf{V}_k(\mathbf{x})$.

Moreover, we have

$$
\begin{aligned}
&\mathbb{E}\|I_{12}\| \\
\le& \frac{1}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\mathbb{E} \left\|\frac{1}{n} \sum_{i=1}^{n} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right\|^2\right]^{1/2} \left\|\mathbb{E} \mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0)\right\| \\
\le& \frac{C}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\frac{1}{n^2} \sum_{i=1}^{n} \mathbb{E} \left\|\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E} \frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right\|^2\right]^{1/2} \\
\le& \frac{C}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \left[\frac{1}{n} \mathbb{E} \left\|\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}, \boldsymbol{\theta}_0) \mathbf{V}_k^\top(\mathbf{x})\right\|^2\right]^{1/2} \le \frac{C}{k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} \frac{1}{\sqrt{n}} \sqrt{k} = O(n^{-1/2}) = o(1).
\end{aligned}
$$

For I_2, observe that

$$
\begin{aligned}
\mathbb{E}\|I_2\| \le& \frac{1}{n^2 k_{\max}^{3/2}} \sum_{i=1}^{n} \sum_{k=k_{\min}}^{k_{\max}} \mathbb{E}\|\mathbf{V}_k(\mathbf{x}_i)\|^2 \|\frac{\partial}{\partial \boldsymbol{\theta}} g(\mathbf{x}_i, \boldsymbol{\theta}_0)\|^2 \\
\le& \frac{C}{n k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} (\mathbb{E}\|\mathbf{V}_k(\mathbf{x})\|^4)^{1/2} = \frac{C}{n k_{\max}^{3/2}} \sum_{k=k_{\min}}^{k_{\max}} k = O(n^{-1} k_{\max}^{1/2}).
\end{aligned}
$$

This finishes the proof of $T_2/\sigma_n = o_P(1)$.

To show $T_3/\sigma_n = o_P(1)$, note that

$$\begin{aligned}\frac{1}{\sigma_n}T_3 =&(\boldsymbol{\theta}_0 - \widehat{\boldsymbol{\theta}})^\top \frac{1}{\sigma_n}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\varepsilon_i\\ =&\sqrt{n}(\boldsymbol{\theta}_0 - \widehat{\boldsymbol{\theta}})^\top \frac{1}{\sigma_n}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\varepsilon_i\end{aligned}$$

and $\sqrt{n}(\boldsymbol{\theta}_0 - \widehat{\boldsymbol{\theta}}) = O_P(1)$, so that it suffices to show

$$\begin{aligned}\frac{1}{\sigma_n\sqrt{n}}\tilde{T}_3 =&\frac{1}{n^{3/2}k_{\max}^{3/2}}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\varepsilon_i\\ =&\frac{1}{n^{3/2}k_{\max}^{3/2}}\sum_{i=1}^{n}\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\left[\sum_{j=1,\neq i}^{n}\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right]\varepsilon_i = o_P(1).\end{aligned}$$

In fact, under H_0,

$$\begin{aligned}\mathbb{E}\left\|\frac{1}{\sigma_n\sqrt{n}}\tilde{T}_3\right\|^2 =& \frac{1}{n^3k_{\max}^3}\mathbb{E}\left\|\sum_{i=1}^{n}\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\left[\sum_{j=1,\neq i}^{n}\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right]\varepsilon_i\right\|^2\\
=&\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=1}^{n}\mathbb{E}\left\|\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\left[\sum_{j=1,\neq i}^{n}\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right]\right\|^2\\
=&\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=1}^{n}\mathbb{E}\left\|\sum_{j=1,\neq i}^{n}\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right\|^2\\
=&\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\mathbb{E}\left\|\sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right\|^2\\
&+\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=1}^{n}\sum_{j_1=1,\neq i}^{n}\sum_{j_2=1,\neq i,\neq j_1}^{n}\\
&\mathbb{E}\sum_{k=k_{\min}}^{k_{\max}}(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_{j_1},\boldsymbol{\theta}_0)^\top\mathbf{V}_k^\top(\mathbf{x}_{j_1}))\mathbf{V}_k(\mathbf{x}_i)\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbb{E}(\mathbf{V}_k(\mathbf{x}_{j_2})\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_{j_2},\boldsymbol{\theta}_0))\\
\leq&\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=1}^{n}\sum_{j=1,\neq i}^{n}\mathbb{E}\left(\|\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\|\sum_{k=k_{\min}}^{k_{\max}}|\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)|\right)^2\\
&+\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=1}^{n}\sum_{j_1=1,\neq i}^{n}\sum_{j_2=1,\neq i,\neq j_1}^{n}\\
&\sum_{k_1=k_{\min}}^{k_{\max}}\sum_{k_2=k_{\min}}^{k_{\max}}(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_{j_1},\boldsymbol{\theta}_0)\mathbf{V}_{k_1}^\top(\mathbf{x}_{j_1}))\mathbb{E}[\mathbf{V}_{k_1}(\mathbf{x}_i)\mathbf{V}_{k_2}^\top(\mathbf{x}_i)]\mathbb{E}(\mathbf{V}_{k_2}(\mathbf{x}_{j_2})\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_{j_2},\boldsymbol{\theta}_0))\\
\leq&\frac{c_1}{n}k_{\max} + \frac{c_2}{k_{\max}} = o(1),\end{aligned}$$

where we have used the following results: (1) $\mathbb{E}\left(\sum_{k_1=k_{\min}}^{k_{\max}}|\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)|\right)^4 \leq Ck_{\max}^8$ for some absolutely constant C that is the same order as $\mathbb{E}|a_{ij}|^4$ we used before; (2) $\mathbb{E}[\mathbf{V}_{k_1}(\mathbf{x}_i)\mathbf{V}_{k_2}^\top(\mathbf{x}_i)]$ is a $k_1 \times k_2$ matrix with 1's on the diagonal and all other elements being zero; $\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_{j_1},\boldsymbol{\theta}_0)\mathbf{V}_{k_1}^\top(\mathbf{x}_{j_1})$ and $\mathbb{E}(\mathbf{V}_{k_2}(\mathbf{x}_{j_2})\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_{j_2},\boldsymbol{\theta}_0))$ are matrices of coefficients of projecting $\mathbf{x}_{j_1}$ and $\mathbf{x}_{j_2}$ onto $\mathbf{V}_{k_1}^\top(\mathbf{x}_{j_1})$ and $\mathbf{V}_{k_2}^\top(\mathbf{x}_{j_2})$, so that their norms are bounded by $\mathbb{E}\left\|\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x},\boldsymbol{\theta}_0)\right\|$. The proof is finished. $\square$

Proof of Theorem A.4. Under H_1, $m_n(x) = a_n m(x)$ with $\|m(x)\| > 0$. Our model is written as $y = g(\mathbf{x}, \boldsymbol{\theta}_0) + m_n(\mathbf{x}) + e$ where $\mathbb{E}(e|\mathbf{x}) = 0$. Given a sample $\{(y_i, \mathbf{x}_i), i = 1, 2, \cdots, n\}$ and a truncation parameter k, though $a_n \to 0$, the SP method can still estimate $\boldsymbol{\theta}_0$ consistently from the sample models, because $\mathbf{M}_v$ does not depend on a_n at all. Then we write $\widetilde{e}_i = y_i - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) = e_i + m_n(\mathbf{x}_i) + g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})$.

It follows that

$$
\begin{aligned}
\widehat{\sigma}_e^2 =& \frac{1}{n}\sum_{i=1}^n \widetilde{e}_i^2 = \frac{1}{n}\sum_{i=1}^n [e_i + m_n(\mathbf{x}_i) + g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})]^2 \\
=& \frac{1}{n}\sum_{i=1}^n e_i^2 + a_n^2 \frac{1}{n}\sum_{i=1}^n m(\mathbf{x}_i)^2 + \sum_{i=1}^n [g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})]^2 \\
&+ \frac{2}{n}\sum_{i=1}^n [g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})]^2 e_i + a_n \frac{2}{n}\sum_{i=1}^n m(\mathbf{x}_i) e_i \\
&+ a_n \frac{2}{n}\sum_{i=1}^n [g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}})]^2] m(\mathbf{x}_i) \to_P \mathbb{E}[e^2]
\end{aligned}
$$

as $n \to \infty$, in view of $\boldsymbol{\theta}_0 - \widehat{\boldsymbol{\theta}} = o_P(1)$, the first order Taylor approximation of $g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) = \dfrac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}, \boldsymbol{\theta}_0)(\boldsymbol{\theta}_0 - \widehat{\boldsymbol{\theta}})$, and the i.i.d. data process of $\{\mathbf{x}_i, e_i\}$. Hence, in what follows, we replace $\widehat{\sigma}_e^2$ involved in σ_n^2 by $\mathbb{E}[e^2]$, while noticing again $\sigma_n^2 = O(n^2 k_{\max}^3)$.

Observe that

$$
\begin{aligned}
T_n =& \sum_{i=1}^n \sum_{j=1,\neq i}^n \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\widetilde{e}_i \widetilde{e}_j = 2\sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\widetilde{e}_i \widetilde{e}_j \\
=& 2\sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_i) + e_i] \\
&\quad \times [g(\mathbf{x}_j, \boldsymbol{\theta}_0) - g(\mathbf{x}_j, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_j) + e_j] \\
=& 2\sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j) e_i e_j \\
&+ 2\sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j) e_i [g(\mathbf{x}_j, \boldsymbol{\theta}_0) - g(\mathbf{x}_j, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_j)] \\
&+ 2\sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j) e_j [g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_i)] \\
&+ 2\sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j) [g(\mathbf{x}_i, \boldsymbol{\theta}_0) - g(\mathbf{x}_i, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_i)] \\
&\quad \times [g(\mathbf{x}_j, \boldsymbol{\theta}_0) - g(\mathbf{x}_j, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_j)] \\
=& T_{1n} + T_{2n} + T_{3n} + T_{4n}, \quad \text{say.}
\end{aligned}
$$

Since e_i has the same property as ε_i under H_0, it follows from the proof of Theorem 3.2 that $T_{1n}/\sigma_n \to_{\mathcal{D}} N(0,1)$. To fulfil the proof, we shall then show that T_{2n}/σ_n and T_{3n}/σ_n are both of $O_P(a_n k_{\max}^{1/2})$, while $T_{4n}/\sigma_n \geq O_P(a_n^2 n / k_{\max}^{1/2}) \to_P \infty$.

As a matter of fact, we have

$$
\begin{aligned}
\frac{1}{\sigma_n} T_{2n} =& \frac{1}{\sigma_n} \sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j) e_i [g(\mathbf{x}_j, \boldsymbol{\theta}_0) - g(\mathbf{x}_j, \widehat{\boldsymbol{\theta}}) + m_n(\mathbf{x}_j)] \\
=& \sqrt{n}(\boldsymbol{\theta}_0 - \widehat{\boldsymbol{\theta}})^\top \frac{1}{n^{3/2} k_{\max}^{3/2}} \sum_{i=2}^n \sum_{j=1}^{i-1} \sum_{k=k_{\min}}^{k_{\max}} \mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j) \frac{\partial}{\partial \boldsymbol{\theta}^\top} g(\mathbf{x}_j, \boldsymbol{\theta}_0) e_i
\end{aligned}
$$

$$+\frac{a_n}{nk_{\max}^{3/2}}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)e_i,$$

and

$$\begin{aligned}&\mathbb{E}\left\|\frac{1}{n^{3/2}k_{\max}^{3/2}}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)e_i\right\|^2\\ =&\frac{\sigma_e^2}{n^3k_{\max}^3}\sum_{i=2}^{n}\mathbb{E}\left\|\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\right\|^2=o(1)\end{aligned}$$

in view of the same derivation of $\frac{1}{\sigma_n\sqrt{n}}\tilde{T}_3$ in Theorem 3.2; moreover,

$$\begin{aligned}&\mathbb{E}\left|\frac{a_n}{nk_{\max}^{3/2}}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)e_i\right|^2\\ =&\frac{\sigma_e^2a_n^2}{n^2k_{\max}^3}\sum_{i=2}^{n}\mathbb{E}\left|\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)\right|^2\\ =&\frac{\sigma_e^2a_n^2}{n^2k_{\max}^3}\sum_{i=2}^{n}\mathbb{E}\left|\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\sum_{j=1}^{i-1}\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)\right|^2\\ \le&\frac{\sigma_e^2a_n^2}{n^2k_{\max}^3}\sum_{i=2}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbb{E}\|\mathbf{V}_k^\top(\mathbf{x}_i)\|^2\sum_{k=k_{\min}}^{k_{\max}}\mathbb{E}\left\|\sum_{j=1}^{i-1}\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)\right\|^2\\ =&\frac{Ca_n^2}{n^2k_{\max}}\sum_{i=2}^{n}\sum_{k=k_{\min}}^{k_{\max}}\mathbb{E}\left\|\sum_{j=1}^{i-1}[\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)-\mathbb{E}\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)+\mathbb{E}\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)]\right\|^2\\ =&\frac{Ca_n^2}{n^2k_{\max}}\sum_{i=2}^{n}\sum_{k=k_{\min}}^{k_{\max}}\left[\sum_{j=1}^{i-1}\mathbb{E}\|\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)-\mathbb{E}\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)\|^2+\|\mathbb{E}\mathbf{V}_k(\mathbf{x}_j)m(\mathbf{x}_j)\|^2\right]\\ \le&\frac{Ca_n^2}{k_{\max}}\sum_{k=k_{\min}}^{k_{\max}}\left[\mathbb{E}\|\mathbf{V}_k(\mathbf{x})m(\mathbf{x})\|^2+\|\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})\|^2\right]\le\frac{Ca_n^2}{k_{\max}}\sum_{k=k_{\min}}^{k_{\max}}k=Ck_{\max}a_n^2,\end{aligned}$$

where C is absolute constant that may be different in each appearance. In addition, it is evident that $\frac{1}{\sigma_n}T_{3n}$ has the same order as $\frac{1}{\sigma_n}T_{2n}$.

Furthermore, we obtain

$$\begin{aligned}&\frac{1}{\sigma_n}T_{4n}\\ =&\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})+m_n(\mathbf{x}_j)][g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})+m_n(\mathbf{x}_i)]\\ =&\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)m_n(\mathbf{x}_j)m_n(\mathbf{x}_i)\\ &+\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})][g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})]\\ &+\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)[g(\mathbf{x}_j,\boldsymbol{\theta}_0)-g(\mathbf{x}_j,\widehat{\boldsymbol{\theta}})]m_n(\mathbf{x}_i)\end{aligned}$$

$$
\begin{aligned}
&+\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)m_n(\mathbf{x}_j)[g(\mathbf{x}_i,\boldsymbol{\theta}_0)-g(\mathbf{x}_i,\widehat{\boldsymbol{\theta}})]\\
=&\frac{1}{\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\left\|\sum_{i=1}^{n}\mathbf{V}_k^\top(\mathbf{x}_i)m_n(\mathbf{x}_i)\right\|^2-\frac{1}{\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\sum_{i=1}^{n}\|\mathbf{V}_k^\top(\mathbf{x}_i)m_n(\mathbf{x}_i)\|^2\\
&+\sqrt{n}(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})^\top\left[\frac{1}{n\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\left(\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\right)\right.\\
&\qquad\left.\times\left(\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\right)^\top\right]\sqrt{n}(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})\\
&-\sqrt{n}(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})^\top\left[\frac{1}{n\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_i)\frac{\partial}{\partial\boldsymbol{\theta}}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\right]\sqrt{n}(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})\\
&+\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})m_n(\mathbf{x}_i)\\
&+\frac{2}{\sigma_n}\sum_{i=2}^{n}\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\mathbf{V}_k^\top(\mathbf{x}_i)\mathbf{V}_k(\mathbf{x}_j)m_n(\mathbf{x}_j)\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)(\boldsymbol{\theta}_0-\widehat{\boldsymbol{\theta}})\\
=&J_1+\cdots+J_6,\quad\text{say.}
\end{aligned}
$$

Notice that

$$
\begin{aligned}
J_1=&\frac{a_n^2}{\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\left\|\sum_{i=1}^{n}\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)\right\|^2=\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left\|\frac{1}{n}\sum_{i=1}^{n}\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)\right\|^2\\
=&\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left\|\mathbb{E}\mathbf{V}_k^\top(\mathbf{x})m(\mathbf{x})+\frac{1}{n}\sum_{i=1}^{n}[\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)]\right\|^2\\
=&\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\|\mathbb{E}\mathbf{V}_k^\top(\mathbf{x})m(\mathbf{x})\|^2+\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left\|\frac{1}{n}\sum_{i=1}^{n}[\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)]\right\|^2\\
&+2\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}[\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})]^\top\left(\frac{1}{n}\sum_{i=1}^{n}[\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)]\right)\\
=&\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\|\tilde{\boldsymbol{\gamma}}\|^2+\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left\|\frac{1}{n}\sum_{i=1}^{n}[\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)]\right\|^2\\
&+2\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}[\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})]^\top\left(\frac{1}{n}\sum_{i=1}^{n}[\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)]\right)\\
\geq&\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}[\mathbb{E}m^2(\mathbf{x})-\sum_{j=k}^{\infty}\tilde{\gamma}_j^2]+\frac{a_n^2n}{k_{\max}^{3/2}}\cdot O_P\left(\sum_{k=k_{\min}}^{k_{\max}}\frac{k}{n}\right)\\
&-2\frac{a_n^2n}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\|\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})\|\left\|\frac{1}{n}\sum_{i=1}^{n}[\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)]\right\|\\
\geq&\|m(x)\|^2\frac{a_n^2n}{k_{\max}^{1/2}}-\frac{a_n^2n}{k_{\max}^{1/2}}\sum_{j=k_{\min}}^{\infty}\gamma_j^2+O_P(a_n^2k_{\max}^{1/2})-2\|m(x)\|\frac{a_n^2n}{k_{\max}^{3/2}}\frac{1}{\sqrt{n}}O_P\left(\sum_{k=k_{\min}}^{k_{\max}}\sqrt{k}\right)\\
=&\|m(x)\|^2\frac{a_n^2n}{k_{\max}^{1/2}}-O_P\left(\frac{a_n^2n}{k_{\max}^{1/2}}\sum_{j=k_{\min}}^{\infty}\gamma_j^2\right)+O_P(a_n^2k_{\max}^{1/2})-O_P(a_n^2\sqrt{n})
\end{aligned}
$$

$$=\frac{a_n^2 n}{k_{\max}^{1/2}}(1+o_P(1)),$$

where $\tilde{\gamma}_j$ are coefficients in the expansion of $m(x)$, $\tilde{\boldsymbol{\gamma}} = (\tilde{\gamma}_1, \cdots, \tilde{\gamma}_k)^\top$ and by Parseval equality $\|m(x)\|^2 = \mathbb{E}[m^2(\mathbf{x})] = \sum_{j=1}^\infty \tilde{\gamma}_j^2$, and the order of $\sum_k \left\| \frac{1}{n}\sum_{i=1}^n [\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i) - \mathbb{E}\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)] \right\|^2 = O_P\left(\sum_{k=k_{\min}}^{k_{\max}} k/n\right)$ is derived straightforwardly so that we omit it.

Next, we analyze $J_2, \cdots, J_6$ term by term. For J_2,

$$\begin{aligned}
J_2 =& \frac{1}{\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\sum_{i=1}^n \|\mathbf{V}_k^\top(\mathbf{x}_i)m_n(\mathbf{x}_i)\|^2 = \frac{a_n^2}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\frac{1}{n}\sum_{i=1}^n \|\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)\|^2 \\
=& \frac{a_n^2}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}} \mathbb{E}\|\mathbf{V}_k^\top(\mathbf{x})m(\mathbf{x})\|^2 + \frac{a_n^2}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{i=1}^n \|\mathbf{V}_k^\top(\mathbf{x}_i)m(\mathbf{x}_i)\|^2 - \mathbb{E}\|\mathbf{V}_k^\top(\mathbf{x})m(\mathbf{x})\|^2\right) \\
=& O(a_n^2 k_{\max}^{1/2}) + \frac{a_n^2}{k_{\max}^{3/2}\sqrt{n}} O_P\left(\sum_{k=k_{\min}}^{k_{\max}} k\right) = O(a_n^2 k_{\max}^{1/2}) + o_P(1).
\end{aligned}$$

For J_3, consider

$$\begin{aligned}
&\frac{1}{n\sigma_n}\sum_{k=k_{\min}}^{k_{\max}}\left(\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\right)\left(\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\right)^\top \\
=& \frac{1}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\right)\left(\frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i)\right)^\top \\
=& \frac{1}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}))(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}))^\top \\
&+ \frac{1}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right) \\
&\quad\times\left(\frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)^\top \\
&+ \frac{1}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}))\left(\frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)^\top \\
&+ \frac{1}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{i=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_i,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_i) - \mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}))^\top \\
\le& \mathbb{E}\|\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\|^2\frac{1}{k_{\max}^{1/2}} + \frac{1}{k_{\max}^{3/2}n}O_P\left(\sum_{k=k_{\min}}^{k_{\max}} k\right) + \frac{2}{k_{\max}^{3/2}\sqrt{n}}O_P\left(\sum_{k=k_{\min}}^{k_{\max}}\sqrt{k}\right) = o_P(1),
\end{aligned}$$

where $\left\|\left(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)\left(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)^\top\right\| \le \mathbb{E}\left\|\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\right\|^2$ since $\mathbb{E}\left[\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right]$ are these coefficients of $\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)$ projecting onto $\mathbf{V}_k^\top(\mathbf{x})$. Thus, $J_3 = o_P(1)$. This implies $J_4 = o_P(1)$.

For J_5, observe

$$\begin{aligned}
&\frac{2}{\sqrt{n}\sigma_n}\sum_{i=2}^n\sum_{j=1}^{i-1}\sum_{k=k_{\min}}^{k_{\max}}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_j)\mathbf{V}_k(\mathbf{x}_i)m_n(\mathbf{x}_i) \\
=& \frac{a_n\sqrt{n}}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{j=1}^n \frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_j)\right)\left(\frac{1}{n}\sum_{i=1}^n \mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)\right)
\end{aligned}$$

$$
\begin{aligned}
=&\frac{a_n\sqrt{n}}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}))(\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x}))\\
&+\frac{a_n\sqrt{n}}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}(\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}))\left(\frac{1}{n}\sum_{i=1}^n\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})\right)\\
&+\frac{a_n\sqrt{n}}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{j=1}^n\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_j)-\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})\\
&+\frac{a_n\sqrt{n}}{k_{\max}^{3/2}}\sum_{k=k_{\min}}^{k_{\max}}\left(\frac{1}{n}\sum_{j=1}^n\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x}_j,\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x}_j)-\mathbb{E}\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right)\\
&\qquad\times\left(\frac{1}{n}\sum_{i=1}^n\mathbf{V}_k(\mathbf{x}_i)m(\mathbf{x}_i)-\mathbb{E}\mathbf{V}_k(\mathbf{x})m(\mathbf{x})\right)\\
=&O\left(\frac{a_n\sqrt{n}}{k_{\max}^{1/2}}\right)+2\frac{a_n\sqrt{n}}{k_{\max}^{3/2}}O_P\left(\sum_{k=k_{\min}}^{k_{\max}}\sqrt{k/n}\right)+\frac{a_n\sqrt{n}}{k_{\max}^{3/2}}O_P\left(\sum_{k=k_{\min}}^{k_{\max}}k/n\right)=O_P\left(a_n\sqrt{n/k_{\max}}\right),
\end{aligned}
$$

where once again $\left\|\mathbb{E}\left[\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\mathbf{V}_k^\top(\mathbf{x})\right]\right\|\leq\sqrt{\mathbb{E}\left\|\frac{\partial}{\partial\boldsymbol{\theta}^\top}g(\mathbf{x},\boldsymbol{\theta}_0)\right\|^2}$ and $\|\mathbb{E}m(\mathbf{x})\mathbf{V}_k^\top(\mathbf{x})\|\leq\sqrt{\mathbb{E}[m^2(\mathbf{x})]}$. Thus, $J_5=O_P\left(a_n\sqrt{n/k_{\max}}\right)$, and $J_6=O_P\left(a_n\sqrt{n/k_{\max}}\right)$ due to their similarity.

To conclude, $\frac{1}{\sigma_n}T_{4n}\geq a_n^2 n/k_{\max}^{1/2}(1+o_P(1))\to_P\infty$, and hence, the theorem is fulfilled. □

E Additional Simulations

In this appendix, we connect our work with Gallant (1981) and Gao et al. (2002), and present more numerical evidences with the corresponding theoretical justification to support our SP method.

E.1 Generalized cross–validation

Firstly, we introduce an alternative method for selecting the truncation parameter when sparsity assumptions are not ideal for practical analysis. We emphasize that the primary objective is not the specific selection of k, but rather the execution of a rigorous robustness check for our SP method.

Specifically, we focus on the so–called "Generalized Cross–Validation" (GCV) method (see, Chapter 2 of Härdle et al. (2000), for example) works well numerically. For the linear model discussed in Section 2, the GCV selection method is defined as follows:

$$
\widehat{k}_n=\underset{k\in K_n}{\operatorname{argmin}}\left(1-\frac{k+1}{n}\right)^{-2}\cdot\frac{1}{n}\sum_{i=1}^n\left(y_i-\mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}-\widehat{m}_\phi(\mathbf{x}_i)\right)^2, \tag{E.1}
$$

where $\widehat{m}_\phi(\mathbf{x}_i)=\sum_{j=1}^k\phi_j(\mathbf{x}_i)\,\widehat{\gamma}_{j,\phi}$ with $\phi_j(\cdot)$ being an orthonormal sequence of basis functions chosen by the user from $L^2([0,\pi])$ and $\widehat{\gamma}_{j,\phi}$ being defined in the same way as in Section 3, in which $K_n=\left\{1,2,\cdots,\left[c_1\,n^{\frac{1}{5}}\right]\right\}$, in which c_1 is the user choice in an individual scenario such that K_n can have up to certain integers. An asymptotic consistency for $\widehat{k}_n$ can be established in a similar way to Gao et al. (2002).

E.2 Simulated examples

Example E1: Consider the following data generating process for a univariate setting:

$$
y_i=x_i\,\beta_0+\varepsilon_i\quad\text{with}\quad\varepsilon_i=m(x_i)+e_i,
$$

where $e_i\sim N(0,1)$, $\beta_0=1$, and $i=1,\ldots,n$. Additionally, we consider the following cases for x_i's and $m(x)$:

Case A: $x_i \sim U(-2,1)$, and $m(x) = 15\cos(3\pi x)$;

Case B: $x_i \sim N(1,1)$, and $m(x) = x^2 - 2$;

Case C: $x_i \sim U(-1,2)$, and $m(x) = \exp(x) - x - \frac{\exp(2)-\exp(-1)}{3} + \frac{1}{2}$;

Case D: Consider Case A, and add an extra exogenous variable to the model, so $y_i = x_i\,\beta_0 + w_i\,\alpha_0 + \varepsilon_i$, where $w_i \sim N(1,1)$ and $\alpha_0 = 1$;

In Appendix E.2 below, we show that Cases A–D all fulfil the following endogeneity condition:

$$\mathbb{E}[m(x_i)] = 0 \quad \text{and} \quad \mathbb{E}[x_i\, m(x_i)] \neq 0. \tag{E.2}$$

In what follows, we provide some detailed implementation and make comments on each case. We start with Cases A–C.

1. **LS**–Estimate β_0 by the OLS method:

$$\widehat{\beta}_{\mathrm{LS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y},$$

   where we let $\mathbf{X} = (x_1,\ldots,x_n)^\top$ and $\mathbf{y} = (y_1,\ldots,y_n)^\top$ for notational simplicity.

2. **SP**–Estimate β_0 by the SP method of the form:

$$\widehat{\beta}_{\mathrm{SP}} = (\mathbf{X}^\top\mathbf{M}_v\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{M}_v\mathbf{y}),$$

   where $\mathbf{M}_v = \mathbf{I}_n - \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$, and $\mathbf{V} = (\mathbf{v}(x_1),\ldots,\mathbf{v}(x_n))^\top$. Here, we take the suggestions from Gallant (1981, p. 228), and choose $\mathbf{v}(x)$ as $\mathbf{v}(x) = (1, H_2(x), p_1(x), p_2(x), \ldots, p_{k-2}(x))^\top$. An optimal choice of k as $\widehat{k}_n$ is selected by the GCV method defined in (E.1).

In practice, one can further add more polynomials to $\mathbf{v}(x)$ if necessary. The above setting is already sufficient to evaluate our theory, so we no longer explore higher order polynomials. We repeat the above procedure $R = 1000$ times, and report the following measures

$$\Delta\beta = \overline{\beta} - \beta_0, \qquad \mathrm{sd}_\beta = \left\{\frac{1}{R}\sum_{r=1}^{R}(\widehat{\beta}_r - \overline{\beta})^2\right\}^{1/2}, \quad \overline{k} = \frac{1}{R}\sum_{r=1}^{R}\widehat{k}_{n,r}, \tag{E.3}$$

where $\overline{\beta} = \frac{1}{R}\sum_{r=1}^{R}\widehat{\beta}_r$ with $\widehat{\beta}_r \in \{\widehat{\beta}_{\mathrm{LS}}, \widehat{\beta}_{\mathrm{SP}}\}$ in each replication, $\widehat{k}_{n,r}$ stands for the value of $\widehat{k}_n$ chosen by the GCV in the r^{th} replication, and $\overline{k}$ reported in the following tables represents the largest integer part.

For Case D, we also consider LS and SP, but choose

$$\mathbf{X} = \begin{pmatrix} x_1 & \ldots & x_n \\ w_1 & \ldots & w_n \end{pmatrix}^\top.$$

The rest setting is identical to those specified for Cases A–C. On top of the above measures in (E.3), we report $\Delta\alpha$ and sd_α which are calculated in the same way as $\Delta\beta$ and sd_β.

We summarize the results in Table 1, and provide the key findings below.

1. Across all scenarios, the SP method consistently delivers strong performance. In contrast, the LS method consistently demonstrates noticeable bias and, in some instances, exhibits considerable standard deviation.

Table 1: Results of Cases A-D of Example E1

		Case A			Case C				
	n	$\Delta\beta$	sd_β	$\overline{k}$	$\Delta\beta$	sd_β			$\overline{k}$
LS	200	-0.091	0.757		0.782	0.101			
	300	-0.126	0.599		0.784	0.080			
	400	-0.107	0.533		0.783	0.067			
SP	200	0.004	0.137	5	0.067	0.131			4
	300	-0.007	0.107	5	0.063	0.109			4
	400	0.006	0.098	5	0.070	0.096			4
		Case B			Case D				
	n	$\Delta\beta$	sd_β	$\overline{k}$	$\Delta\beta$	sd_β	$\Delta\alpha$	sd_α	$\overline{k}$
LS	200	0.981	0.160		-0.187	0.838	-0.062	0.584	
	300	0.992	0.125		-0.163	0.664	-0.039	0.473	
	400	0.992	0.110		-0.154	0.562	-0.040	0.402	
SP	200	-0.003	0.129	2	-0.003	0.135	-0.001	0.072	5
	300	-0.005	0.104	2	0.004	0.110	0.003	0.058	5
	400	0.002	0.086	2	-0.003	0.094	0.001	0.051	5

2. In Case C, simple algebra shows that $\exp(x) - x = \sum_{j=0}^{\infty} \frac{x^j}{j!} - x$, so the linear term disappear from the right hand side. While truncation residuals are inherent in this scenario, the proposed SP method still achieves significantly lower bias and smaller standard deviation.

3. For Case D, the proposed SP method surpasses the LS method by yielding unbiased estimates and reduced standard deviations for both parameters.

4. The optimal value of k is case-dependent. Specifically, for optimal finite sample performance, we anticipate (i) $\widehat{k}_n \geq 5$ in Cases A and D, and (ii) $\widehat{k}_n \geq 2$ for Cases B and C. These expectations align with the results presented in Table 1. Furthermore, the value of k suggests that a slightly over-specified k still yields reasonable finite sample performance. This is a positive finding, as it indicates that the choice of k is quite robust in practice.

We now have a look at a bivariate case for $\mathbf{x} = (x_1, x_2)^\top$ below.

Example E2: Consider the following DGP for a bivariate setting:

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i \quad \text{and} \quad \varepsilon_i = m(\mathbf{x}_i) + e_i, \tag{E.4}$$

where $\mathbf{x}_i = (x_{1i}, x_{2i})^\top$, $\boldsymbol{\beta}_0 = (\beta_{01}, \beta_{02})^\top = (1,1)^\top$, and $e_i \sim N(0,1)$ for $1 \leq i \leq n$.

As discussed in the end of Appendix E.2 below, the following conditions can be verified:

$$\mathbb{E}[m(\mathbf{x}_i)] = 0 \quad \text{and} \quad \mathbb{E}[\mathbf{x}_i\, m(\mathbf{x}_i)] \neq \mathbf{0}. \tag{E.5}$$

We start with two additive cases:

Case A: $m(\mathbf{x}) = x_1^2 + x_2^2 - 13/3$, $x_{1i} \sim U(-2,3)$, and $x_{2i} \sim N(1,1)$;

Case B: $m(\mathbf{x}) = 15\cos(3\pi x_1) + x_2^2 - 2$, $x_{1i} \sim U(-2,1)$, and $x_{2i} \sim N(1,1)$.

Accordingly, we let

$$\mathbf{v}(\mathbf{x}) = (1, H_2(x_1), p_1(x_1), \ldots, p_{k-2}(x_1), H_2(x_2), p_1(x_2), \ldots, p_{k-2}(x_2))^\top,$$

where $k \geq 2$ ensures the length of $\mathbf{v}(\mathbf{x})$ is at least 3. Apparently, in this case, $\mathbf{v}(\mathbf{x})$ is $(2k-1)\times 1$. An optimal choice of k as $\widehat{k}_n$ is selected by the GCV method:

$$\widehat{k}_n = \underset{k \in K_n}{\operatorname{argmin}} \left(1 - \frac{2k}{n}\right)^{-2} \cdot \frac{1}{n} (\mathbf{y} - \mathbf{X}\widehat{\beta}_{\mathrm{SP}} - \mathbf{V}\widehat{\boldsymbol{\gamma}})^\top (\mathbf{y} - \mathbf{X}\widehat{\beta}_{\mathrm{SP}} - \mathbf{V}\widehat{\boldsymbol{\gamma}}),$$

Table 2: Results of Bivariate Case of Example E2

		Case A					Case B				
	n	$\Delta\beta_1$	sd_{β_1}	$\Delta\beta_2$	sd_{β_2}	$\overline{k}$	$\Delta\beta_1$	sd_{β_1}	$\Delta\beta_2$	sd_{β_2}	$\overline{k}$
LS	200	0.712	0.128	0.809	0.192		0.409	0.792	1.099	0.622	
	300	0.714	0.110	0.811	0.164		0.405	0.667	1.109	0.484	
	400	0.715	0.092	0.809	0.142		0.422	0.593	1.119	0.421	
SP	200	0.000	0.062	-0.002	0.128	3	0.003	0.130	0.002	0.130	6
	300	-0.001	0.052	-0.004	0.099	3	0.000	0.110	0.007	0.101	6
	400	0.000	0.043	0.000	0.091	3	0.003	0.101	-0.001	0.088	6
		Case C					Case D				
LS	200	-0.554	0.472	2.414	0.607		-0.467	3.953	-0.158	4.543	
	300	-0.522	0.376	2.452	0.477		-0.402	3.021	-0.017	3.701	
	400	-0.517	0.332	2.458	0.444		-0.614	2.567	-0.117	3.023	
SP	200	-0.001	0.062	-0.006	0.134	3	-0.005	0.151	-0.011	0.136	6
	300	0.002	0.054	-0.001	0.102	3	0.000	0.117	-0.006	0.104	6
	400	0.001	0.043	0.000	0.087	3	0.002	0.098	-0.002	0.091	6

where $\widehat{\gamma} = (\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top(\mathbf{y} - \mathbf{X}\widehat{\beta}_{\text{SP}})$, and K_n is as defined in (E.1).

Meanwhile, we also consider the following multiplicative cases:

Case C: $m(\mathbf{x}) = x_1^2 \cdot (x_2^2 - 2)$, $x_{1i} \sim U(-2, 3)$, and $x_{2i} \sim N(1, 1)$;

Case D: $m(\mathbf{x}) = 15\cos(3\pi x_1) \cdot (x_2^2 + 2)$, $x_{1i} \sim U(-2, 1)$, and $x_{2i} \sim N(1, 1)$.

Accordingly, we let $\mathbf{v}(\mathbf{x})$ include the distinct elements from

$$\{1, H_2(x_1), p_1(x_1), \ldots, p_{k-2}(x_1)\} \otimes \{1, H_2(x_2), p_1(x_2), \ldots, p_{k-2}(x_2)\},$$

where $k \geq 2$ ensures the length of $\mathbf{v}(\mathbf{x})$ is at least 4. An optimal choice of k as $\widehat{k}_n$ is selected by the GCV method:

$$\widehat{k}_n = \arg\min_{k \in K_n} \left(1 - \frac{k^2+1}{n}\right)^{-2} \cdot \frac{1}{n}(\mathbf{y} - \mathbf{X}\widehat{\beta}_{\text{SP}} - \mathbf{V}\widehat{\boldsymbol{\gamma}})^\top(\mathbf{y} - \mathbf{X}\widehat{\beta}_{\text{SP}} - \mathbf{V}\widehat{\boldsymbol{\gamma}}), \tag{E.6}$$

where the quantities involved are the same as those in (E.1).

For the bivariate settings, the findings in Table 2 support that the proposed SP method works well numerically in a very similar spirit to that of Table 1 for Example E1.

Example E3: We now examine the nonlinear models. Without loss of generality, we consider the model: $y_i = g(\mathbf{x}_i, \boldsymbol{\theta}_0) + \varepsilon_i$.

Case A: $g(x, \theta_0) = \frac{1}{1+\exp(-x\theta_0)}$, $m(x) = 0.5\cos(3\pi x)$, and $x_i \sim U(-\frac{13}{6}, \frac{7}{6})$, where x_i and $\theta_0\,(= -1)$ are scalars.

Case B: $g(x, \theta_0) = \sin(x\theta_0)$, $m(x) = 5(x^2 - 3)$, and $x_i \sim U(-2, 1)$, where x_i and $\theta_0\,(= -1)$ are scalars.

Case C: $g(\mathbf{x}, \boldsymbol{\theta}_0) = \theta_4 \exp(x_1\theta_1) + \theta_5 \exp(\mathbf{x}^\top\boldsymbol{\theta}_{23})$, where $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5) = (-1, -0.5, -0.5, 1, 1)$, $\boldsymbol{\theta}_{23} = (\theta_2, \theta_3)^\top$, and $\mathbf{x} = (x_1, x_2)^\top$. Additionally, we consider the case where $m(x_1, x_2) = \mathbb{E}[\varepsilon|(x_1, x_2)] = \mathbb{E}[\varepsilon|x_1] = m(x_1) = 6\cos(3\pi x_1)$, $x_{1,i} \sim U(-\frac{13}{6}, \frac{7}{6})$, and $x_{2,i} \sim N(1, 1)$.

As shown for the justification of Example E2 below, endogeneity exists in all three cases. The forms of $m(\cdot)$ in the above cases are chosen similarly to those in Cases A and B of Example E1.

In Case A, we select $g(x, \theta_0)$ as a sigmoid function (i.e., logistic function), which receives lots of attention in the optimization process of machine learning literature (e.g., Bauer and Kohler, 2019).

In Case B, we select $g(x, \theta_0)$ as a standard periodic function, which is possibly one of the easiest nonlinear functions that one can think of.

Table 3: Results of Nonlinear Models of Example E3

		Case A				Case B		
	n	$\Delta\theta$	sd_θ	$\overline{k}$		$\Delta\theta$	sd_θ	$\overline{k}$
LS	200	-0.351	0.741			2.102	0.917	
	300	-0.305	0.444			2.233	0.693	
	400	-0.289	0.366			2.252	0.682	
SP	200	-0.099	0.802	5		-0.022	0.178	3
	300	-0.056	0.504	5		-0.007	0.139	3
	400	-0.045	0.428	5		-0.010	0.122	3
				Case C				
		$\Delta\theta_1$	sd_{θ_1}	$\Delta\theta_2$	sd_{θ_2}	$\Delta\theta_3$	sd_{θ_3}	$\overline{k}$
LS	1500	3.375	3.268	0.003	0.101	0.035	0.155	
	2000	3.786	3.223	0.000	0.084	0.053	0.138	
	2500	4.007	3.241	-0.003	0.077	0.057	0.131	
SP	1500	0.005	0.155	-0.003	0.033	0.001	0.024	5
	2000	0.003	0.136	-0.001	0.030	0.001	0.021	5
	2500	0.002	0.116	0.000	0.027	0.000	0.019	5
		$\Delta\theta_4$	sd_{θ_4}	$\Delta\theta_5$	sd_{θ_5}	$\overline{k}$		
LS	1500	-0.527	1.079	0.414	0.985			
	2000	-0.547	0.788	0.429	0.686			
	2500	-0.566	0.656	0.426	0.540			
SP	1500	0.033	0.233	0.004	0.078	5		
	2000	0.028	0.212	0.004	0.065	5		
	2500	0.018	0.174	0.005	0.058	5		

In Case C, the functional form is similar to those typically adopted in the neural network literature, and the exponential function is the fundamental component of a series of activation functions such as Softmax, Softplus, Gaussian function, logistic function, etc.

To carry on the regression, we construct $\mathbf{M}_v$ of Case A and Case B as in Example E1, construct $\mathbf{M}_v$ of Case C as in Example E2, and finally obtain the estimates for all three cases as in (A.14). The optimal value of k is still selected via GCV with obvious modification. For the purpose of comparison, we also carry on the standard nonlinear least squares (NLS) method, and report $\Delta\theta$, sd_θ and $\overline{k}$ as defined in (E.3).

In Table 3, the proposed SP method is apparently dominating the traditional NLS method. The endogeneity causes some serious bias issues for the NLS method. As more unknown parameters are involved in Case C, we therefore consider larger sample sizes in the simulation. In Case C, the endogeneity is due to x_1 only, and it apparently influences the estimates of these coefficients in different magnitude. The values of optimal k are similar to those in Cases A and B of Example E1. As discussed above, these values should be expected.

E.3 Verifications of the simulation designs in Appendix E.1

In what follows, x_i stands for a random variable, while x stands for a real number.

Lemma E.1. $\{\cos(i\pi x) \mid i \in 0 \cup \mathbb{N}\}$ *is a set of orthogonal basis functions in* $L^2([a, b])$ *for* $\forall a, b \in \mathbb{Z}$ *and* $a \neq b$.

Proof of Lemma E.1.

For $i \neq j$, write

$$\int_a^b \cos(i\pi x)\cos(j\pi x)\mathrm{d}x = \frac{1}{2}\int_a^b \cos((i-j)\pi x)\mathrm{d}x + \frac{1}{2}\int_a^b \cos((i+j)\pi x)\mathrm{d}x$$
$$= \frac{1}{2(i-j)\pi}\sin((i-j)\pi x)\Big|_a^b + \frac{1}{2(i+j)\pi}\sin((i+j)\pi x)\Big|_a^b = 0,$$

where the last step is obvious.

Also, we write

$$\int_a^b \cos(i\pi x)^2 \mathrm{d}x = \frac{1}{2}\int_a^b \cos(0\pi x)\mathrm{d}x + \frac{1}{2}\int_a^b \cos(2i\pi x)\mathrm{d}x$$
$$= \frac{b-a}{2} + \begin{cases} \frac{b-a}{2} & \text{for } i = 0 \\ \frac{1}{4i\pi}\sin(2i\pi x)|_a^b & \text{for } i \geq 1 \end{cases} = \begin{cases} b-a & \text{for } i = 0 \\ \frac{b-a}{2} & \text{for } i \geq 1 \end{cases}.$$

The proof is completed. □

Justification of Example E1.

We consider the cases one by one.

Case A: Let ℓ be an odd positive integer. We note that

$$\mathbb{E}[\cos(\ell\pi x_i)] = \int_{-2}^{1} \frac{1}{3}\cos(\ell\pi x)\mathrm{d}x = \frac{1}{9\pi}\sin(\ell\pi x)\big|_{-2}^{1} = 0.$$

Also,

$$\mathbb{E}[x_i\cos(\ell\pi x_i)] = \int_{-2}^{1} \frac{1}{3}x\cdot\cos(\ell\pi x)\mathrm{d}x = \frac{1}{3\ell\pi}\int_{-2}^{1} x\,\mathrm{d}\sin(\ell\pi x)$$
$$= \frac{1}{3\ell\pi}x\cdot\sin(\ell\pi x)\big|_{-2}^{1} - \frac{1}{3\ell\pi}\int_{-2}^{1}\sin(\ell\pi x)\mathrm{d}x = \frac{1}{3(\ell\pi)^2}\cos(\ell\pi x)\big|_{-2}^{1} = \frac{-2}{3(\ell\pi)^2},$$

and

$$\mathbb{E}[x_i^2\cos(\ell\pi x_i)] = \int_{-2}^{1} \frac{1}{3}x^2\cdot\cos(\ell\pi x)\mathrm{d}x = \frac{1}{3\ell\pi}\int_{-2}^{1} x^2\,\mathrm{d}\sin(\ell\pi x)$$
$$= -\frac{2}{3\ell\pi}\int_{-2}^{1} x\cdot\sin(\ell\pi x)\mathrm{d}x = \frac{2}{3(\ell\pi)^2}\int_{-2}^{1} x\,\mathrm{d}\cos(\ell\pi x)$$
$$= \frac{2}{3(\ell\pi)^2}x\cdot\cos(\ell\pi x)\big|_{-2}^{1} - \frac{2}{3(\ell\pi)^2}\int_{-2}^{1}\cos(\ell\pi x)\mathrm{d}x = \frac{2}{3(\ell\pi)^2}.$$

Case B: We note that if $x_i \sim N(\mu, 1)$, then $\mathbb{E}[H_j(x_i)] = \mu^j$. The reason is as follows.

$$\mathbb{E}[H_j(x_i)] = \int_{\mathbb{R}} H_j(x)f(x-\mu)\mathrm{d}x = \int_{\mathbb{R}} H_j(x+\mu)f(x)\mathrm{d}x$$
$$= \int_{\mathbb{R}}\sum_{k=0}^{j}\binom{j}{k}\mu^{j-k}H_j(x)f(x)H_0(x)\mathrm{d}x = \mu^j\int_{\mathbb{R}} H_0(x)^2 f(x)\mathrm{d}x = \mu^j,$$

where $f(x)$ is the density function of $N(0,1)$, $H_0(x) \equiv 1$, and the fourth equality follows from the orthogonality of $\{H_j(x)\}$. We them immediately obtain that

$$\mathbb{E}[x_i^2 - 2] = \mathbb{E}[H_2(x_i) - H_0(x_i)] = \mu^2 - \mu^0,$$

and $\mathbb{E}[x_i(x_i^2-2)] = \mathbb{E}[H_3(x_i) + H_1(x_i)] = \mu^3 + \mu^1$. Thus, for $\mu = 1$, $\mathbb{E}[x_i^2 - 2] = 0$ and $\mathbb{E}[x_i(x_i^2-2)] = 2$.

Case C: We note that $\exp(x) - x = \sum_{j=0}^{\infty}\frac{x^j}{j!} - x$, so the liner term disappear from the right hand side. Simple algebra shows that

$$\int_{-1}^{2}\frac{1}{3}(\exp(x) - x)\mathrm{d}x = \frac{1}{3}\exp(x)\big|_{-1}^{2} - \frac{1}{6}x^2\big|_{-1}^{2} = \frac{1}{3}\exp(2) - \frac{1}{3}\exp(-1) - \frac{1}{2}.$$

Thus, $\mathbb{E}[m(x)] = 0$.

We also have

$$\int_{-1}^{2}\frac{1}{3}x\exp(x)\mathrm{d}x = \frac{1}{3}x\exp(x)\big|_{-1}^{2} - \int_{-1}^{2}\frac{1}{3}\exp(x)\mathrm{d}x$$

$$= \frac{2}{3}\exp(2) + \frac{1}{3}\exp(-1) - \frac{1}{3}\exp(2) + \frac{1}{3}\exp(-1) = \frac{1}{3}\exp(2) + \frac{2}{3}\exp(-1)$$

and $\int_{-1}^{2} \frac{1}{3}x^2 \mathrm{d}x = \frac{1}{9}x^3\big|_{-1}^{2} = 1$. Therefore, we have

$$\mathbb{E}[x\, m(x)] = \frac{1}{3}\exp(2) + \frac{2}{3}\exp(-1) - 1 - \frac{1}{3}\exp(2) + \frac{1}{3}\exp(-1) + \frac{1}{2} = \exp(-1) - \frac{1}{2} \neq 0.$$

Remark: We note that, for Cases A, C and D, it is easy to see that $\mathbb{E}[\mathbf{v}(x_j)\mathbf{v}(x_j)^\top]$ is not diagonal but has full rank based on the above development. For Case B, it is mathematically more involved, as we need to use Euler's formula: $\cos(x) = \frac{1}{2}(\exp(\mathrm{i}x) + \exp(-\mathrm{i}x))$, where i is an imagine unit. Thus,

$$\begin{aligned}\mathbb{E}[\cos(x_j)] &= \frac{1}{2}(\mathbb{E}[\exp(\mathrm{i}x_j)] + \mathbb{E}[\exp(-\mathrm{i}x_j)]) \\ &= \frac{1}{2}\exp\left(\mathrm{i}\mu_x - \frac{\sigma_x^2}{2}\right) + \frac{1}{2}\exp\left(-\mathrm{i}\mu_x - \frac{\sigma_x^2}{2}\right) = \exp\left(-\frac{\sigma_x^2}{2}\right)\cos(\mu_x),\end{aligned}$$

where μ_x and σ_x are the mean and standard deviation of the random variable x_j, and the second equality follows from a direct calculation using the characteristic function of a normal distribution. As this is simulation exercise, we no longer purse the final form of $\mathbb{E}[\mathbf{v}(x_j)\mathbf{v}(x_j)^\top]$ further, which also varies with respect to the choice of the Hermite function(s) involved. □

Justification of Example E2.

We now show that all cases fulfil (E.5).

In Example E2, Cases A and B can be easily justified by following the justification of Cases A and B of Example E1, as $m(\mathbf{x})$ admits an additive form: $m(\mathbf{x}) = m_1(x_1) + m_2(x_2)$ in both cases.

In Cases C and D, $m(\mathbf{x})$ always admits an multiplicative form: $m(\mathbf{x}) = m_1(x_1) \cdot m_2(x_2)$. In connection with the justification of Cases A and B of Example E1, we can also verify (E.5). □

Justification of Example E3.

Case A: Note that given $\theta_0 < 0$, $\frac{1}{1+\exp(-x\theta_0)}$ is monotonically decreasing with respect to x, and $\cos(3\pi x)$ is a periodic function with respect to x. It is easy to check that $\int_{-\frac{13}{6}}^{\frac{7}{6}} \cos(3\pi x)\mathrm{d}x \int = \frac{1}{3\pi}\int_{-\frac{13}{2}\pi}^{\frac{7}{2}\pi} \cos(x)\mathrm{d}x = 0$ and

$$\begin{aligned}&\int_{-\frac{13}{6}}^{\frac{7}{6}} \frac{1}{1+\exp(-x\theta_0)}\cos(3\pi x)\mathrm{d}x \\ &> \sum_{k=-5,-3,-1,1,3} \frac{1}{1+\exp((2k-1)(-\theta_0)/6)}\left(\int_{(2k-3)/6}^{(2k-1)/6} + \int_{(2k-1)/6}^{(2k+1)/6}\right)\cos(3\pi x)\mathrm{d}x = 0.\end{aligned}$$

Thus, Case A of Example E3 fulfils the condition $E[g(x_i, \theta_0)m(x_i)] \neq 0$.

Case B: By direct calculation, we immediately obtain that $\int_{-2}^{1} m(x)\mathrm{d}x = 0$, (i.e., $E[m(x_i)] = 0$). The level of endogeneity can be calculated as follows:

$$\begin{aligned}&\int_{-2}^{1} \sin(-x)\cdot 5(x^2-3)\mathrm{d}x = -5\int_{-2}^{1}\sin(x)x^2\mathrm{d}x + 5\int_{-2}^{1}\sin(x)\mathrm{d}x \\ &= -4\cos(1) + 7\cos(2) + 2\sin(1) - 4\sin(2) \approx -7.\end{aligned}$$

Thus, Case B of Example E3 fulfils the condition $E[g(x_i, \theta_0)m(x_i)] \neq 0$.

Case C: In view of the development of Case A, the justification of Case C is obvious, as $\exp(-x_1)$ is also monotonically decreasing. □

References

Bauer, B. and Kohler, M. (2019). On deep learning as a remedy for the curse of dimensionality in nonparametric regression. *Annals of Statistics*, 47(4):2261–2285.

Dong, C. and Gao, J. (2025). *Modern Series Methods in Econometrics and Statistics.* Springer, New York.

Dong, C., Linton, O., and Peng, B. (2021). A weighted sieve estimator for nonparametric time series models with nonstationary variables. *Journal of Econometrics*, 222(3):909–932.

Gallant, A. R. (1981). On the bias in flexible functional forms and an essentially unbiased form: The Fourier flexible form. *Journal of Econometrics*, 15(2):211–245.

Gao, J., Tong, H., and Wolff, R. (2002). Adaptive orthogonal series estimation in additive stochastic regression models. *Statistica Sinica*, 11(4):409–428.

Hall, P. G. and Heyde, C. C. (1980). *Martingale Limit Theory and Its Applications.* Academic Press, New York.

Härdle, W., Liang, H., and Gao, J. (2000). *Partially Linear Models.* Springer/Verlag, New York.

Park, J. Y. and Phillips, P. C. B. (2001). Nonlinear regression with integreted time series. *Econometrica*, 69(1):117–161.